

Lesson 01 Demo 02

Applying CoT and ReAct Prompting for Drafting User Stories and Analyzing Problems

Objective: To demonstrate how Chain-of-Thought (CoT) and ReAct prompting techniques can help product managers reason through user stories and product problems and simulate decision-making using structured thinking and tool-based action planning

Tools required: ChatGPT (or any LLM interface)

Prerequisites: Basic understanding of user stories, product discovery, and reasoning frameworks (CoT and ReAct)

Steps to be followed:

1. Break down a user story using CoT prompting
2. Analyze a product problem using CoT prompting
3. Simulate market research using ReAct prompting

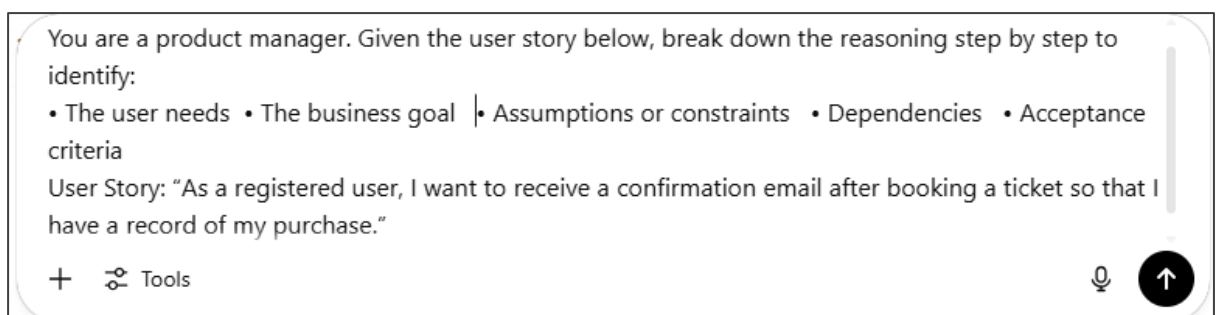
Step 1: Break down a user story using CoT prompting

1.1 Open ChatGPT and enter the following prompt:

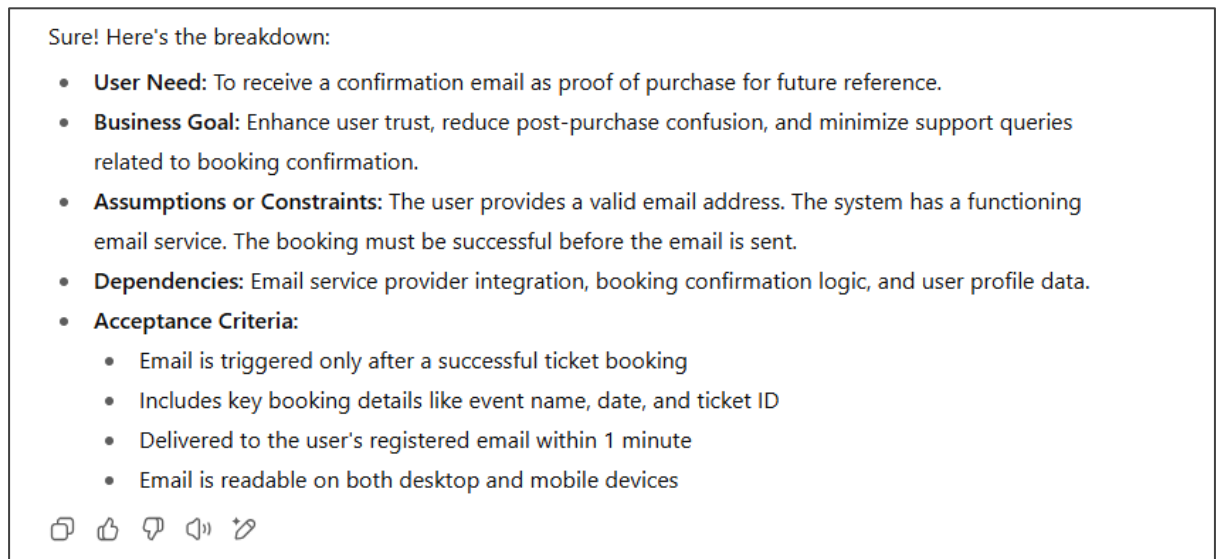
You are a product manager. Given the user story below, break down the reasoning step by step to identify:

- The user needs
- The business goal
- Assumptions or constraints
- Dependencies
- Acceptance criteria

User Story: “As a registered user, I want to receive a confirmation email after booking a ticket so that I have a record of my purchase.”



Expected output may look as shown in the following screenshot:



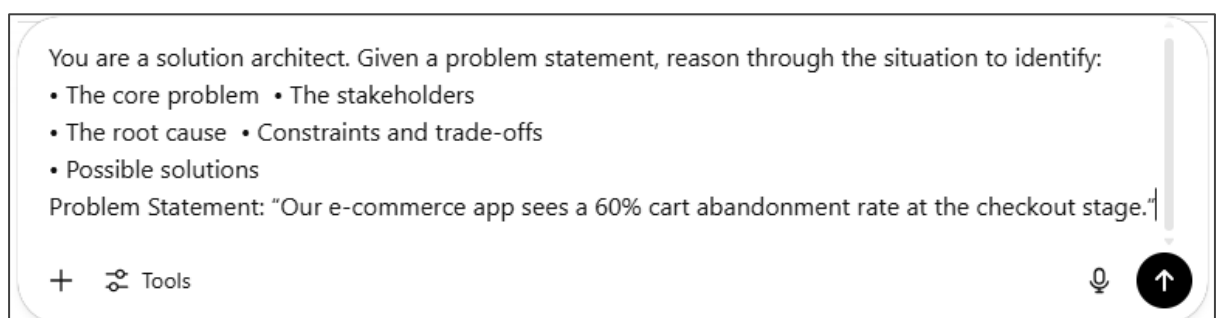
Step 2: Analyze a product problem using CoT prompting

2.1 Now, enter the following prompt:

You are a solution architect. Given a problem statement, reason through the situation to identify:

- The core problem
- The stakeholders
- The root cause
- Constraints and trade-offs
- Possible solutions

Problem Statement: "Our e-commerce app sees a 60% cart abandonment rate at the checkout stage."



The expected output may look as shown in the following screenshot:

Sure! Here's the reasoning breakdown:

- **Core Problem:** A high number of users are dropping off during the checkout process, resulting in lost sales.
- **Stakeholders:** End users (shoppers), product management team, engineering team, marketing team, and business leadership.
- **Root Cause:** Possible friction in the checkout flow—such as mandatory account creation, poor mobile optimization, unexpected costs (like shipping or taxes), or limited payment options.
- **Constraints and Trade-offs:**
 - Balancing security and speed in payment flow
 - Third-party payment gateway limitations
 - Ensuring compliance (e.g., PCI, GDPR)
 - Maintaining data accuracy while reducing user effort

- **Possible Solutions:**
 - Enable guest checkout
 - Simplify the form and reduce required fields
 - Add a progress indicator during checkout
 - Show total costs (including shipping) earlier
 - Offer multiple payment options including digital wallets



Step 3: Simulate market research using ReAct prompting

3.1 Enter the following prompt, using ReAct style:

You are a product manager researching AI-based writing assistants. Think step by step. Use search when needed to get competitor information. Then summarize findings into 3 user story ideas.

Available Tools: [Search], [Thought], [Write]

Begin:

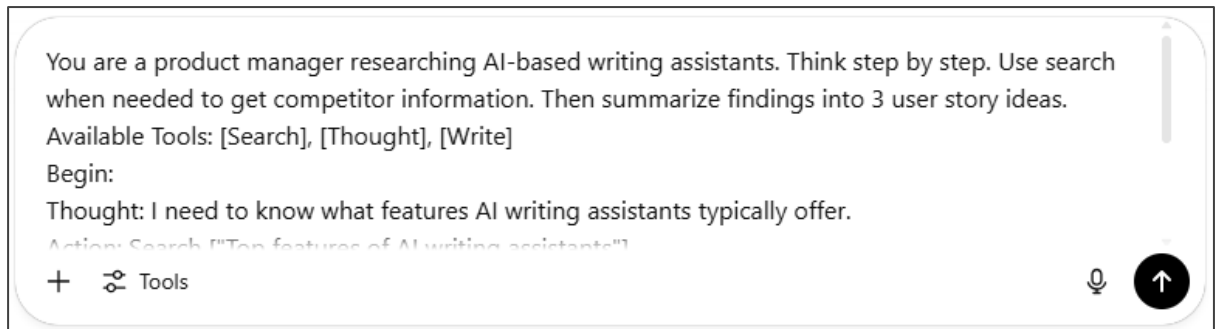
Thought: I need to know what features AI writing assistants typically offer.

Action: Search ["Top features of AI writing assistants"]

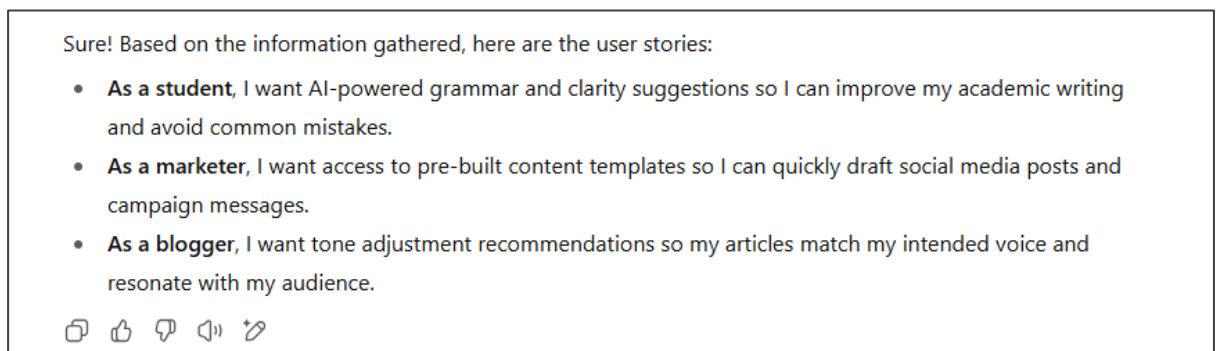
Observation: Most AI writing tools include grammar checks, tone suggestions, rewriting, and content templates.

Thought: These features could be helpful for students, marketers, and bloggers.

Action: Write ["Create 3 user stories based on this information."]



The expected output may look as shown in the following screenshot:



By following these steps, you will be able to apply Chain-of-Thought and ReAct prompting to break down user stories, analyze product problems, and perform market research, whether clarifying acceptance criteria or generating targeted user stories for students, marketers, and bloggers.